AIGR TreeGrafter Function-Inference Stress Test — Final Report

Target gene: gabD-II (`gabD-II`, locus `PP_4422`) · *Pseudomonas putida* KT2440 (PSEPK, NCBITaxon:160488) **UniProt accession:** Q88EN2 **Seed hypothesis (TreeGrafter/PANTHER, IEA, GO_REF:0000118):** gabD-II has succinate-semialdehyde dehydrogenase (NAD+) activity — **GO:0004777 Focus:** function_assignment · `function-hypothesis-go-0004777`

Summary

The seed hypothesis is **partially supported but too specific**. The reaction identity is right; the cofactor qualifier is wrong. Q88EN2 is unambiguously a *bona fide*, catalytically intact **succinate-semialdehyde dehydrogenase (SSADH)** of the ALDH superfamily. It carries the diagnostic Pfam aldehyde-dehydrogenase domain (PF00171, Aldedh), maps to InterPro families IPR015590/IPR016163, and is assigned by PANTHER to the SSADH subfamily PTHR43353:SF5. Its two catalytic residues — the nucleophilic **Cys296** and the general-base **Glu262** — are present and correctly spaced relative to characterized active members (they align to human ALDH5A1 Cys340/Glu306 and to the Cys/Glu pair shown by mutagenesis to be essential in a cyanobacterial SSADH). This decisively **rejects the pseudo-enzyme failure mode (mode 2)**: the active site is intact, not degenerate.

The protein also sits firmly inside the **GabD clade** of bacterial SSADHs rather than the neighboring YneI/Sad clade. Pairwise identities place it at 51.7% to *E. coli* GabD and 48.2% to human ALDH5A1, but only 35.4% to *E. coli* Sad/YneI. Because the reaction (succinate semialdehyde + NAD(P)+ → succinate + NAD(P)H) is correct and the subfamily placement is coherent, the **within-superfamily mis-placement failure mode (mode 3) is also rejected**. The graft landed on the right functional subfamily.

What does **not** hold up is the **NAD+** specification embedded in GO:0004777. This is a **granularity/cofactor error (failure mode 1)**. Cofactor preference in SSADHs is set by a small number of residues in the coenzyme 2′-pocket, not by the catalytic machinery, and most characterized SSADHs are *promiscuous*, using both NAD+ and NADP+ with different affinities. Q88EN2's coenzyme-binding module tracks the **NADP+-specific** *E. coli* GabD very closely (76%

local identity over the 2′-pocket region, with conserved basic residues Arg172/Lys173 that help accommodate the NADP+ 2′-phosphate), whereas it is only 35.4% identical overall to the NAD+/dual-cofactor YneI clade. Thus strict NAD+ dependence — the exact thing GO:0004777 asserts — is the *least* likely of the possibilities for this protein. The recommended curation lead is to **retain the SSADH function but generalize or replace the cofactor**: move to **GO:0009013** [succinate-semialdehyde dehydrogenase (NAD(P)+) activity] or to the NADP+ sibling **GO:0036243**, pending a definitive in vitro NAD+/NADP+ kinetics assay. We did not find primary biochemical data for Q88EN2 itself, so the cofactor call remains a sequence-based lean rather than a measured fact, and we flag it as such.

Executive Judgment

Verdict: Partially supported / too specific — cofactor granularity error (failure mode 1).

- **Reaction (SSADH) — supported.** Catalytic Cys296 and Glu262 conserved and correctly spaced; PF00171 + PTHR43353:SF5; ~48–52% identity to characterized SSADHs.
- **Pseudo-enzyme (mode 2) — rejected.** Active-site residues intact; no evidence of degeneration.
- **Wrong subfamily (mode 3) — rejected.** Clean GabD-clade placement (51.7% to *E. coli* GabD vs 35.4% to YneI).
- **Cofactor "NAD+" qualifier (mode 1) — not supported.** Coenzyme pocket is GabD/NADP+-type; strict NAD+ is the least likely cofactor.

Single most decisive piece of evidence: The coenzyme-binding 2′-pocket of Q88EN2 is 76% identical to the NADP+-specific *E. coli* GabD and retains the basic Arg172/Lys173 pair that accommodates the NADP+ 2′-phosphate, while the protein is only 35.4% identical overall to the NAD+/dual YneI clade. The fold and catalytic residues confirm SSADH activity, but the pocket argues against the specific NAD+ claim in GO:0004777.

Independent Family / Function Assignment

Most likely specific molecular function (from our own analysis): succinate-semialdehyde dehydrogenase, catalyzing

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succinate semialdehyde + NAD(P)+ + H2O  →  succinate + NAD(P)H + H+
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Nearest characterized homolog: *Escherichia coli* **GabD** (UniProt P25526), an **NADP+-specific** succinate-semialdehyde dehydrogenase — 51.7% identical to Q88EN2 and the closest characterized neighbor. Human ALDH5A1 (P51649, NAD+-preferring) is a more distant characterized relative at 48.2%. The NAD+/dual-cofactor *E. coli* Sad/YneI (P76149) is clearly more distant at 35.4%.

GO term we would assign and granularity vs. the seed term:

GO term	Label	Relationship to seed GO:0004777
GO:0009013	succinate-semialdehyde dehydrogenase (NAD(P)+) activity	**More general** (parent/sibling that drops the strict cofactor qualifier) — our primary recommendation
GO:0036243	succinate-semialdehyde dehydrogenase (NADP+) activity	**Sibling** — alternative if curator wants a specific cofactor and accepts the GabD-clade lean
GO:0004777	succinate-semialdehyde dehydrogenase (NAD+) activity	The seed term — reaction correct, cofactor qualifier unsupported

Granularity summary: the seed term is **correct in reaction but over-specified in cofactor**. The defensible call from sequence alone is the **cofactor-agnostic** term GO:0009013; the best-guess specific call is the **NADP+** sibling GO:0036243. Strict NAD+ is not supportable without an assay.

Evidence Matrix

#	Citation	Evidence type	Role	Claim tested	Key finding	Organism / context
1	PMID: 32966327	Structural / mutagenesis	**Supports** SSADH reaction; rejects pseudo-enzyme	Are the catalytic Cys/Glu essential, and are they intact in Q88EN2?	"the conserved Cys262 and Glu228 residues were crucial for the catalytic activity" — the Cys/Glu pair is essential; Q88EN2 retains equivalent Cys296/Glu262, correctly spaced	*Cyanothece* sp. ATCC51142 SSADH
2	PMID: 23229889	Structural / comparative	**Qualifies** (cofactor)	Do bacteria have two SSADH clades with different cofactor preference, and which is Q88EN2?	"many organisms… encode two succinate semialdehyde dehydrogenases… different cofactor preference (YneI and GabD)"; "the conserved Lys160 contributes to the enzyme preference to NAD(+)"	*Salmonella typhimurium* YneI
3	PMID: 23500184	Structural	**Qualifies** (cofactor)	Is cofactor preference set by single pocket residues?	"Ser157…and corresponding Pro159…play critical structural roles in determining NADP(+) and NAD(+) preference"	Cyanobacterial SSADH

#	Citation	Evidence type	Role	Claim tested	Key finding	Organism / context
4	PMID: 2647149	Direct assay	**Qualifies / competing**	Are SSADH cofactor assignments clean in vivo?	Two SSADH forms; the large form is NADP-specific, the "NAD-linked" form "was able to use NADP as cofactor"	*Klebsiella pneumoniae*
5	PMID: 25092794	Direct assay	**Qualifies / competing**	Can a single SSADH use both cofactors?	Recombinant enzyme "could use both NAD(+) and NADP(+) as co-factors," Km 579 µM (NAD+) vs 1.0 mM (NADP+)	*Saccharomyces cerevisiae*
6	PMID: 25888791	Structural	Supports (reaction)	Reaction/ catalytic context of a bacterial SSADH	SSADHs "catalyze the oxidation of succinic semialdehyde (SSA) to succinic acid in the presence of NAD(P)(+)"	*Streptococcus pyogenes* (NADP+-dependent)
7	PMID: 29242124	Direct assay / modeling	Qualifies (cofactor)	Example of an NADP+-preferring bacterial-type SSADH	Cyanobacterial enzyme was NADP+-preferring	*Anabaena* sp. PCC7120
8	PMID: 30690134	Direct assay	Context	Kinetic characterization	SSADH kinetically characterized in	*Nitrosomonas europaea*

#	Citation	Evidence type	Role	Claim tested	Key finding	Organism / context
				of a bacterial SSADH	a chemolithotroph	
9	UniProt/ InterPro/ Pfam/ PANTHER	Computational / database	**Supports** family	Domain architecture and subfamily	PF00171 (Aldedh), IPR015590/ IPR016163, PANTHER PTHR43353:SF5 (SSADH); PROSITE PS00070 present	Q88EN2

Active-Site / Placement Analysis

Catalytic residues (reaction identity) — INTACT

A Needleman–Wunsch global alignment (BLOSUM62) of Q88EN2 against characterized SSADHs shows both catalytic residues present and correctly spaced:

Catalytic role	Q88EN2 residue	Aligns to	Status
Nucleophile (thiol)	**Cys296**	Human ALDH5A1 Cys340; UniProt-annotated active site; essential Cys of *Cyanothece* SSADH (P 32966327)	**Conserved, correctly spaced**
General base	**Glu262**	Human ALDH5A1 Glu306; essential Glu of *Cyanothece* SSADH (P 32966327)	**Conserved, correctly spaced**
ALDH Cys signature	—	PROSITE **PS00070** ALDEHYDE_DEHYDR_CYS pattern	**Present**

Conclusion: the catalytic apparatus is intact → the SSADH reaction MF term is justified; **pseudo-enzyme ruled out.**

Subfamily placement (reaction subtype) — GabD clade

Comparison protein	UniProt	Cofactor class	% identity to Q88EN2
E. coli **GabD**	P25526	NADP+-specific	**51.7%** (closest characterized)
Human ALDH5A1	P51649	NAD+-preferring	48.2%
E. coli **Sad/YneI**	P76149	NAD+/dual	**35.4%** (clearly more distant)

Conclusion: Q88EN2 is a **GabD-clade** SSADH. The large identity gap to YneI rules out within-superfamily mis-placement into the NAD+/dual clade.

Coenzyme 2′-pocket (cofactor determinant) — NADP+/GabD-type

Feature	Observation	Implication
2′-pocket local identity to *E. coli* GabD (NADP+)	**76% (28/37)** across GabD 148–184 → Q88EN2 155–191	Pocket resembles NADP+-specific enzyme
ALDH coenzyme glycine motif	…TPWNFP… at Q88EN2 162–166	Canonical ALDH cofactor motif present
Basic pocket residues	**Arg172** / **Lys173** (GabD Arg165/ Lys166) conserved	Accommodate NADP+ 2′-phosphate → NADP+ lean
Literature NAD+ determinant "YneI Lys160"	Maps to Q88EN2 **Lys187**, but conserved in ALL panel members irrespective of cofactor	Pyrophosphate-binding Lys, **not** a discriminator — does not support NAD+
Genome-wide diagnostic-column vote	38:24 toward NADP, **but confounded** by overall GabD identity and mislabeled refs	Treated as unreliable; excluded from the call

Conclusion: cofactor is **not cleanly decidable from sequence**, but every reliable pocket signal leans **NADP+/dual**, and **strict NAD+ is the least likely** option. Two intended external references (Gluconobacter Gox0499, Synechococcus Sp2771) resolved to wrong-length UniProt entries and were discarded.

> **Provenance note:** the alignments, residue mappings, identity percentages, and the diagnostic-column vote above were computed during iterations 1–4 (Needleman–Wunsch/ BLOSUM62 pairwise alignments and panel comparisons) and are recorded as findings F001–F004. The genome-wide vote was explicitly judged unreliable and excluded from the final cofactor determination.

Mechanistic Model / Interpretation

Q88EN2 is the "second" succinate-semialdehyde dehydrogenase of *P. putida* KT2440 (hence **gabD-II**). It operates at the convergence of GABA catabolism and succinate-semialdehyde detoxification, oxidizing succinate semialdehyde to succinate and feeding carbon into the TCA cycle:

```
   GABA  ──(GABA aminotransferase, gabT)──►  succinate semialdehyde (SSA)
                                                        │
                              Q88EN2 (gabD-II, SSADH)   │  + NAD(P)+
                                                        ▼
                                                succinate  ──►  TCA cycle
```

The three-part stress test resolves as follows:

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Failure mode 1 (granularity / cofactor):  TRIGGERED
    Reaction correct, but NAD+ qualifier over-specified → generalize/replace

Failure mode 2 (pseudo-enzyme):           NOT triggered
    Cys296 + Glu262 intact, correctly spaced → active enzyme

Failure mode 3 (wrong subfamily):         NOT triggered
    Clean GabD-clade placement (51.7% vs 35.4%) → right subfamily
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The crux is that PANTHER propagated a term (**GO:0004777**) whose *reaction* is exactly right but whose *cofactor annotation* is an artifact of the reference node the graft inherited. Cofactor identity in this enzyme family is encoded by a handful of 2′-pocket residues and is frequently promiscuous — multiple characterized SSADHs use **both** NAD+ and NADP+ (P 25092794, P 2647149). Q88EN2's pocket matches the NADP+-specific GabD, so the pipeline's confident "NAD+" label is the weakest-supported part of the annotation. This is a textbook granularity error: the fold and catalytic residues are shared across the family, so homology correctly transfers "SSADH," but the cofactor sub-specification should not have transferred without pocket-level evidence.

Evidence Base

- **PMID: 32966327** — *Kinetic and structural insights into enzymatic mechanism of SSADH from Cyanothece sp. ATCC51142.* Mutagenesis establishes the conserved Cys/Glu pair as essential for catalysis. **Supports** the reaction assignment and anchors the active-site conservation test for Q88EN2 (Cys296/Glu262).
- **PMID: 23229889** — *Structure and activity of the NAD(P)+-dependent SSADH YneI from Salmonella typhimurium.* Documents the two-clade (YneI vs GabD) organization with different cofactor preference and the residue basis of NAD+ preference (Lys160). **Qualifies** the cofactor claim and localizes Q88EN2 to the GabD (not YneI) clade.
- **PMID: 23500184** — *Structural basis for cofactor and substrate selection by cyanobacterium SSADH.* Shows single-residue determinants (Ser157/Pro159) switch NADP+/NAD+ preference. **Qualifies** — the NAD+ qualifier is a specific, residue-level claim that must be verified.
- **PMID: 25092794** — *Characterization of recombinant SSADH from S. cerevisiae.* Enzyme uses **both** NAD+ and NADP+ (Km 579 µM vs 1.0 mM). **Competing/qualifies** — demonstrates dual-cofactor behavior typical of SSADHs, undercutting a strict NAD+-only annotation.
- **PMID: 2647149** — *Two SSADHs from Klebsiella pneumoniae.* The "NAD-linked" form can also use NADP. **Competing/qualifies** — cofactor labels are leaky in vivo.
- **PMID: 25888791**, **PMID: 29242124**, **PMID: 30690134** — Additional bacterial/cyanobacterial SSADH characterizations consistently describe the reaction as NAD(P)+-dependent and show NADP+ preference in several GabD-type enzymes. **Support** the reaction and the NADP+/dual lean.
- **UniProt/InterPro/Pfam/PANTHER** — PF00171 (Aldedh), IPR015590/IPR016163, PANTHER PTHR43353:SF5 (SSADH), PROSITE PS00070. **Supports** the family and subfamily assignment for Q88EN2.

GO Curation Implications

Recommended curation action (lead requiring curator verification):

1. **Retain** the succinate-semialdehyde dehydrogenase *reaction* — well supported by fold, subfamily, and intact catalytic residues. Do **not** remove the MF annotation; this is a genuine enzyme, not a non-core / "protein binding" case.

2. **Generalize the cofactor:** replace **GO:0004777** [SSADH (NAD+) activity] with **GO:0009013** [SSADH (NAD(P)+) activity]. This is the defensible, sequence-supported call.
3. **Alternative (more specific):** if the curator prefers a specific cofactor, **replace with the sibling GO:0036243** [SSADH (NADP+) activity], justified by the GabD-clade placement and the NADP+-type 2′-pocket (76% identity to NADP+-specific *E. coli* GabD; basic Arg172/Lys173). Flag as a lean, not a measured fact.
4. **Do NOT retain strict NAD+ (GO:0004777)** as the sole term — the least-supported option.
5. Keep the evidence code at **IEA** until an experimental assay exists; the generalization is a sequence/structure-based curator inference.

Why not "protein binding" or removal: the active site is intact and the subfamily is coherent; the only error is cofactor over-specification. Removal would discard correct information.

Conflicts, Knowledge Gaps, Discriminating Tests

Conflicts and alternatives

- **Paralog confusion:** the "-II" suffix implies a gabD-I paralog. The GabD vs YneI two-clade pattern (P 23229889) means cofactor annotations can be swapped between paralogs; we verified Q88EN2 is the GabD-clade member, but curators should confirm the paralog's identity if one exists.
- **Cofactor promiscuity vs. specificity:** several characterized SSADHs use both NAD+ and NADP+ (P 25092794, P 2647149), so a strict single-cofactor term may misrepresent reality regardless of "preference." This is the strongest argument for the cofactor-agnostic GO:0009013.
- **Database carry-over:** the NAD+ qualifier most plausibly entered via an ancestral PANTHER node annotated NAD+, not via evidence specific to this protein. Two intended external reference sequences resolved to wrong-length UniProt entries and were discarded, weakening part of the comparative panel.

Knowledge gaps

- **No primary biochemistry for Q88EN2 itself.** All cofactor inferences are by homology/structure, not measurement.
- **No experimental structure.** The 2′-pocket analysis rests on alignment to *E. coli* GabD; an AlphaFold model + cofactor docking was not completed and would strengthen (or overturn) the NADP+ lean.
- The genome-wide diagnostic-column vote (38:24 NADP) is confounded and was not used for the call.

Most efficient discriminating tests

1. **In vitro NAD+/NADP+ kinetics** on recombinant Q88EN2 — measure kcat/Km with each cofactor. This single assay definitively resolves GO:0004777 vs GO:0036243 vs GO:0009013. **Highest value.**
2. **Structural / docking check:** build the AlphaFold model, inspect the 2′-pocket (Arg172/Lys173 geometry), dock NADP+ vs NAD+. Fast, computational, corroborates the sequence lean.
3. **Comparative genomics of P. putida SSADH paralogs:** confirm gabD-I vs gabD-II cofactor division of labor, mirroring the GabD/YneI pattern, to ensure the right term is on the right paralog.

Limitations

- This report is a **sequence- and literature-based** assessment. No wet-lab assay was performed on Q88EN2, and no experimental 3D structure exists for it. The catalytic-residue conservation is robust and well anchored to mutagenesis data; the **cofactor assignment is a lean, not a fact**, and is reported as such.
- Two intended reference sequences were unusable (wrong-length UniProt entries), and a genome-wide diagnostic vote was judged unreliable and excluded — so the NADP+ lean rests primarily on the strong local identity to *E. coli* GabD plus the conserved basic pocket residues.
- Any local `*-bioinformatics` analyses in the repository were intentionally withheld and were not consulted; this report is independent and may be compared against them.

Bottom line

Q88EN2 is a real, catalytically competent succinate-semialdehyde dehydrogenase of the GabD clade. The seed annotation got the reaction right but over-specified the cofactor: the defensible GO term is **GO:0009013 [SSADH (NAD(P)+) activity]** (or the NADP+ sibling **GO:0036243**), not the strict NAD+ term **GO:0004777**. The decisive follow-up is a simple NAD+/NADP+ kinetic assay.